Rep Smarter, Not Harder: AI Hypertrophy Coaching with Wearable Sensors and Edge Neural Networks

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***Abstract*—abstract**

I. Introduction

A. Background

Exercise science background

B. Related Work

Literature review

C. Contributions

What we did new

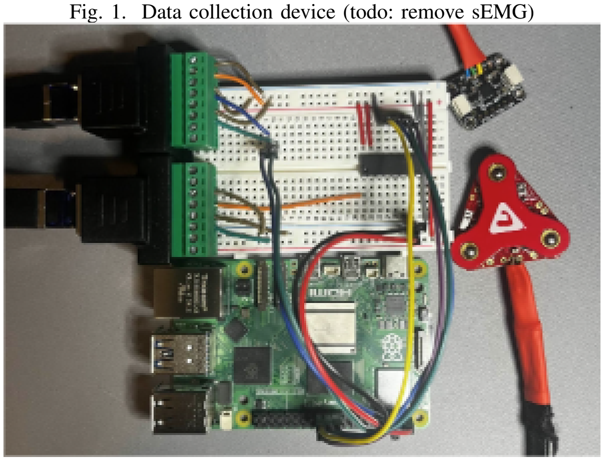
II. Materials and Methods

A. Data Collection

1) Participants: Participants consisted of this work's authors and volunteers at three different gyms. There were (todo) male participants and (todo) female participants, and they varied in weight, height, and resistance training experience.

2) Equipment: Participant preacher curls were recorded using an Adafruit MPU-6050 inertial measurement unit (IMU) on three accelerometer axes and three gyroscope axes. The IMU was connected to a Raspberry Pi 5. The Raspberry Pi was connected to and accessed by a laptop. Three different preacher curl machines at the three locations were used: (todo).

Fig. 1. Data collection device (todo: remove sEMG)



3) Procedure: For each participant, the IMU was taped to their left wrist. Each participant was instructed to perform sets of preacher curls with consistent form and in each set reach muscular failure. Participants were allowed to select the weight they used and how many sets they performed. Consequently, participants varied in the weight they lifted, how many repetitions they performed in each set, and how many sets they volunteered. Participants also varied in their form and range of motion (e.g. locking out their arms at the bottom of the curl, duration of pause at the bottom, height of the concentric phase). Only sets taken to failure with visually consistent form were admitted to this work's dataset.

A Python script on the Raspberry Pi facilitated data collection. The script broadcast an interactive data collection web application for access via laptop. This data collection dashboard was used to start recording a set's IMU data, manually record timestamp markers to annotate the end of each repetition, and end recording.

B. Data Processing

data cleaning, windowing, splitting, augmentation

C. Segmentation Model Design

D. Classification Model Design

E. Model Training

F. Model Evaluation

G. Edge Deployment

describe steps taken to deploy to edge

III. Results

Put any figures and tables here

A. Segmentation Model Performance

B. Classification Model Performance

C. Edge Deployment Performance

IV. Discussion

V. References